

TUDOR POPA

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Hands-on engineering leader with two decades of experience combining software architecture, people management, and platform delivery. Principal Software Engineering Manager at Microsoft, leading Fluent UI React v9 and AI-powered developer tooling. Focused on making AI useful and dependable for developers through retrieval-augmented assistants, MCP tooling, coding-agent workflows, and rigorous evaluation, backed by experience in frontend platforms, distributed systems, and cloud infrastructure.

SKILLS

- **Applied AI & Agents:** LLM applications, Coding agents and skills, Multi-agent workflows, Prompt and context engineering, Model Context Protocol (MCP)
- **Retrieval & Knowledge Systems:** Retrieval-augmented generation (RAG), GraphRAG, Knowledge graphs, Knowledge-base ingestion and indexing
- **LLM & Agent Evaluation:** Promptfoo, LLM-as-judge, Retrieval relevance, Model comparison, Agent benchmarking, CI-backed evaluations
- **Generative UI:** GenUI, Model-first design systems, Headless components, Accessibility, Design-system compliance
- **Programming Languages:** TypeScript, Python, JavaScript, Rust, Go
- **Cloud & Delivery:** Azure, AWS, Google Cloud, Docker, Kubernetes, Terraform, GitHub Actions, Azure Pipelines
- **Databases:** PostgreSQL, MongoDB, Redis, Couchbase
- **Machine Learning Foundations:** TensorFlow, Keras, Brain.js, Predictive modeling, Genetic algorithms

WORK EXPERIENCE

Microsoft December 2023 – Present

Principal Software Engineering Manager (L66) Prague, Czechia

- Promoted from L65 to L66 in December 2023 for driving Fluent UI React v9 adoption and launching the Fluent 2 website; now lead a team of 7 engineers in the Office Product Group owning Fluent UI React v9 and AI-powered developer tooling, including technical strategy, delivery, hiring, and engineer development.
- Lead Fluent Agent RAG platform over curated Fluent UI knowledge bases — scraping docs, Storybook, repos, and llms.txt into FAISS indexes served via OpenAI-compatible chat API and MCP server; deterministic MCP tools validate components, icons, tokens, and props to ground coding agents.
- Authored reusable coding-agent instructions and skills for Fluent UI development and review; added human-in-the-loop issue triage and ownership-aware reviewer assignment.
- Established CI-backed evaluations with Promptfoo, relevance assertions, and LLM-as-judge scoring, plus GenUI benchmarks for API correctness, accessibility, browser behavior, and design-system compliance; used findings to shape headless design-system strategy separating accessible behavior from styling.

Microsoft January 2022 – December 2023

Principal Software Engineering Manager (L65) Prague, Czechia

- Led Fluent UI React v9 platform modernization and adoption, including React 18/19 support, server rendering, migration guidance, documentation, and recurring engineering office hours.
- Led adoption of Astro for the Fluent 2 website; the team delivered 200+ pages in 10 months and halved new-page development time (<https://astro.build/case-studies/microsoft/>).

Getir April 2021 – December 2021

Engineering Manager Cluj-Napoca, Romania

- Defined strategic roadmap for a team of 12 engineers, leading cross-functional collaboration and mentoring junior engineers.
- Managed and grew squads in the Algorithms Tribe, Infrastructure Alliance.
- Led infrastructure algorithms squads, delivering major initiatives to make real-time data feel real-time.
- Empowered and unblocked squads to deliver impact by establishing efficient execution and operational processes.

- Participated in and led software architecture decisions.
- Collaborated with other squads and tribes on cross-functional feature delivery.
- Planned and implemented monitoring systems to prevent service degradations and maintain high throughput.
- Guided team members in career growth, from onboarding to promotions, fostering long-term development.
- Led teams responsible for core business features, including order assignment and courier management.

Black Tree Software Sep 2012 – April 2021

Chief Technology Officer & Co-Owner Cluj-Napoca, Cluj, Romania

- Defined strategic roadmap for a team of 15 engineers, led cross-functional collaboration, and provided mentorship to junior engineers.
- Managed and led multiple engineering teams on various outsourcing projects.
- Drove the development of a sports betting and online gambling white-label PaaS with operations in three African countries (Ghana, Uganda, and Kenya).
- Designed and implemented the software architecture of the whole platform, creating a decoupled, highly scalable, micro-services based system.
- Worked alongside a DevOps team on designing and implementing a complete continuous integration / continuous delivery solution using the AWS cloud.
- Implemented machine learning data models for predicting and categorizing user activities / behaviors.
- Developed a lottery extraction tool which allowed the client to set a custom RTP (return to player) percentage using genetic algorithms.
- Worked with technologies ranging from low level code designed for currency detectors (ccTalk protocol) to full stack solutions (NodeJS, Python and React).

HiPPONET Sep 2012 – Sep 2016

Chief Technology Officer Cluj-Napoca, Cluj, Romania

- Managed a team of 6 developers on various outsourcing projects.
- Collaborated with Swedish company Penvision, developing a SaaS platform for online storage and processing of handwritten documents using proprietary handwriting recognition.
- Re-architected Rahmqvist's CRM tool for the Swedish market.
- Developed and published the Rahmqvist PIMNotes app for both Android and iOS

Wixi.com Sep 2008 – Sep 2012

Lead Engineer / Local Manager Cluj-Napoca, Cluj, Romania

- Managed the Cluj-Napoca office operations.
- Managed a team of 5 engineers building a file hosting service competing directly with Dropbox.
- Architected and implemented a full UI library with comparable window manager-like features in vanilla JS

OpenData Mar 2008 – Jul 2008

Software Engineer Cluj-Napoca, Cluj, Romania

- Designed and implemented a CRM tool for Cluj's largest real estate agency.

WebHorizon Apr 2006 – Jul 2006

Junior Software Engineer Cluj-Napoca, Cluj, Romania

- Worked on the Audi France website, developing components for both the client-facing site and back-office.

EDUCATION

Babeş-Bolyai University Oct 2006 – May 2009

BS Computer Science, Mathematics Cluj-Napoca, Cluj, Romania

Horea, Cloşca şi Crişan High School Sep 2002 – Jun 2006

Informatics, Mathematics Alba-Iulia, Alba, Romania

SIDE PROJECTS

AgentRig TypeScript, MCP, LLM evaluation <https://github.com/doidor/agentrig>

Built a CLI for sharing agent instructions and MCP configurations across coding tools, with fixture-based behavioral evaluations, separate producer/judge conversations, and baseline comparisons.

Markbook TypeScript, React, Vite, Agent skills <https://github.com/doidor/markbook>

Built a Markdown-first documentation and React component showcase tool with Vite, reusable coding-agent skills, and synchronized HTML and llms.txt outputs from a single source.

Github profile Javascript, Python, Bash <https://github.com/doidor>

Various open source projects that I personally use.

Random maze generator Javascript

A personal project that's designed to feed a multiplayer, mobile game with random generated maps, using a genetic algorithm and ranked by an A-star algorithm score in conjunction with a complexity index.